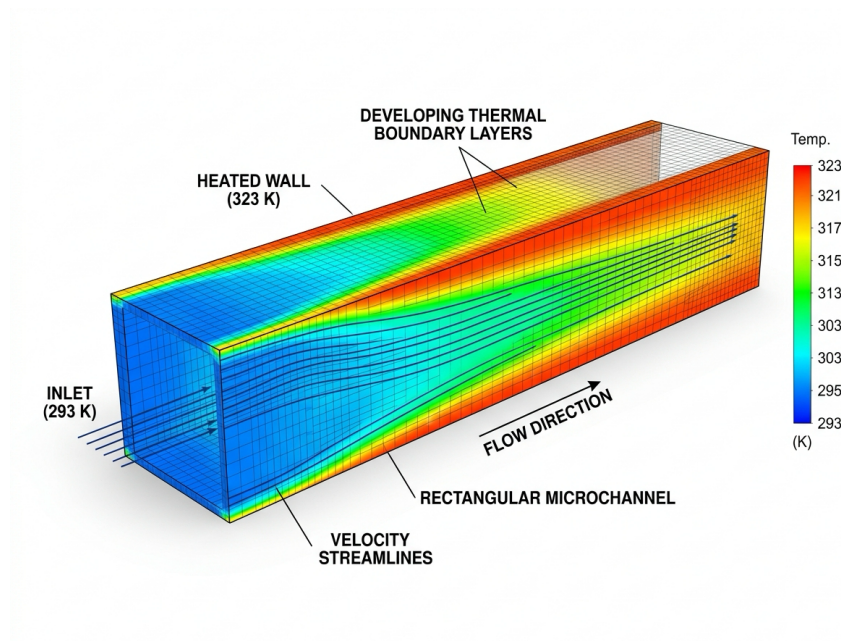


NUMERICAL INVESTIGATION OF LAMINAR FORCED CONVECTION AND HYDRODYNAMIC DEVELOPMENT IN A RECTANGULAR MICROCHANNEL

Three-Dimensional CFD Simulation, Mesh Independence Analysis, and Theoretical Validation Using ANSYS Fluent



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Abstract

This academic report presents a comprehensive three-dimensional numerical investigation of laminar forced convection and hydrodynamic boundary layer development within a rectangular microchannel utilizing ANSYS Fluent. The computational geometry features a width of $W = 0.2$ mm, height of $H = 0.1$ mm (aspect ratio $W/H = 2$), and an axial length of $L = 10$ mm, yielding an equivalent hydraulic diameter of $D_h = 133.3 \mu\text{m}$. A systematic grid independence analysis was performed across five progressively refined computational meshes, spanning from 291,348 to 704,627 elements, to evaluate residual convergence, wall temperature, pressure drop, and the convective heat transfer coefficient. Mesh 1 (704,627 elements) demonstrated asymptotic numerical convergence and was chosen for all subsequent analyses.

Parametric simulations were conducted for Reynolds numbers $Re \in \{100, 200, 300\}$, examining the coupled velocity magnitude and static temperature distributions across longitudinal mid-planes ($y/L_y = 0.5$) and cross-sectional planes ($x/L_x = 0.375$), as well as axial centerline temperature distributions. As the Reynolds number increased from 100 to 300, the convective heat transfer coefficient increased from $2139.44 \text{ W}/(\text{m}^2 \cdot \text{K})$ to $3023.79 \text{ W}/(\text{m}^2 \cdot \text{K})$, and the average wall temperature dropped from 534.05 K to 448.48 K . Numerical results were validated against classical analytical correlations from Kays and London / Incropera for fully developed laminar duct flow under uniform heat flux conditions. The deviations between analytical and numerical Nusselt numbers are rigorously explained by bulk temperature averaging techniques and the pronounced influence of the thermal developing entrance region. Finally, the hydrodynamic entrance length was determined numerically from axial velocity contours and found to be approximately 2.0 mm ($\sim 20\%$ of channel length), in excellent quantitative agreement with the analytical entrance boundary layer relation $L_{fd,h} \approx 0.05 \cdot Re \cdot D_h$.

Keywords: ANSYS Fluent, Microchannel Heat Transfer, Laminar Flow, Grid Independence Study, Nusselt Number, Hydrodynamic Entrance Length, Boundary Layer Development.

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Nomenclature

Symbol	Description	SI Unit
A_c	Duct Cross-Sectional Area ($H \times W$)	m^2
A_b	Duct Base / Heated Wall Area	m^2
D_h	Hydraulic Diameter ($4A_c/P$)	m
h	Convective Heat Transfer Coefficient	$\text{W}/(\text{m}^2 \cdot \text{K})$
H	Channel Height (0.1 mm)	m
k_f	Fluid Thermal Conductivity	$\text{W}/(\text{m} \cdot \text{K})$
L	Channel Axial Length (10 mm)	m
$L_{fd,h}$	Hydrodynamic Entry Length	m
Nu	Nusselt Number (hD_h/k_f)	–
P	Wetted Perimeter	m
P_{in}, P_{out}	Inlet and Outlet Static Pressures	Pa
ΔP	Pressure Drop ($P_{in} - P_{out}$)	Pa
q''	Surface Heat Flux	W/m^2
Re	Reynolds Number ($\rho V D_h/\mu$)	–
T_f	Mean / Bulk Fluid Temperature	K
T_w	Mean Channel Wall Temperature	K
V, V_{in}	Mean Inlet Fluid Velocity	m/s
W	Channel Width (0.2 mm)	m
μ	Dynamic Viscosity	$\text{Pa} \cdot \text{s}$
ρ	Fluid Density	kg/m^3

1 Introduction and Channel Specifications

Microchannel cooling systems are widely deployed in modern electronics thermal management, micro-heat exchangers, and aerospace avionics due to their compact footprint and high surface-area-to-volume ratio, which yields exceptionally high convective heat transfer coefficients. In this study, laminar forced convection of liquid water flowing through a three-dimensional rectangular microchannel is investigated using the finite-volume commercial code ANSYS Fluent.

1.1 Channel Geometry

The rectangular conduit features a height of $H = 0.1 \text{ mm} = 100 \text{ }\mu\text{m}$, a width of $W = 0.2 \text{ mm} = 200 \text{ }\mu\text{m}$, and an axial length of $L = 10 \text{ mm} = 0.01 \text{ m}$. The geometric aspect ratio is $\alpha = W/H = 2$.

The primary geometric parameters are calculated as follows:

$$P = 2(H + W) = 2(0.0001 \text{ m} + 0.0002 \text{ m}) = 6 \times 10^{-4} \text{ m} \quad (1)$$

$$A_c = H \cdot W = 0.0001 \text{ m} \times 0.0002 \text{ m} = 2 \times 10^{-8} \text{ m}^2 \quad (2)$$

$$D_h = \frac{4A_c}{P} = \frac{4 \times (2 \times 10^{-8} \text{ m}^2)}{6 \times 10^{-4} \text{ m}} = 1.333 \times 10^{-4} \text{ m} = 133.3 \text{ }\mu\text{m} \quad (3)$$

1.2 Working Fluid and Thermophysical Properties

Liquid water is utilized as the working fluid, evaluated at reference ambient temperature ($T_{in} \approx 298.15 \text{ K} - 300 \text{ K}$):

- Density (ρ): 997 kg/m^3
- Dynamic viscosity (μ): $8.9 \times 10^{-4} \text{ Pa} \cdot \text{s}$ ($\text{kg}/(\text{m} \cdot \text{s})$)
- Thermal conductivity (k_f): $0.60 \text{ W}/(\text{m} \cdot \text{K})$

1.3 Inlet Velocity Calculation for Baseline Reynolds Number

The flow regime is laminar ($Re \leq 300 \ll 2300$). For the baseline operating condition of $Re = 300$, the uniform inlet velocity V_{in} is derived from the definition of the Reynolds number:

$$V_{in} = \frac{Re \cdot \mu}{\rho \cdot D_h} = \frac{300 \times (8.9 \times 10^{-4} \text{ Pa} \cdot \text{s})}{997 \text{ kg/m}^3 \times (1.333 \times 10^{-4} \text{ m})} = 2.01 \text{ m/s} \quad (4)$$

For the parametric evaluations at $Re = 200$ and $Re = 100$, the corresponding inlet velocities are:

$$Re = 200 \implies V_{in} = \frac{200}{300} \times 2.01 \text{ m/s} = 1.34 \text{ m/s} \quad (5)$$

$$Re = 100 \implies V_{in} = \frac{100}{300} \times 2.01 \text{ m/s} = 0.67 \text{ m/s} \quad (6)$$

1.4 Boundary Conditions

- **Inlet:** Velocity inlet with specified uniform normal velocity ($V_{in} = 2.01, 1.34, 0.67 \text{ m/s}$) and uniform inlet temperature $T_{in} = 298.15 \text{ K}$.
- **Outlet:** Pressure outlet with zero gauge pressure ($P_{gauge} = 0 \text{ Pa}$).
- **Walls:** No-slip kinematic boundary condition ($\vec{v} = 0$). Constant uniform heat flux boundary condition $q'' = 1.22 \times 10^6 \text{ W/m}^2$ applied at the heated boundary area ($A_b = 4 \times 10^{-6} \text{ m}^2$).

2 Task 1: Mesh Convergence and Grid Independence Analysis

To guarantee that the numerical predictions are free of spatial discretization errors, a rigorous grid independence study was performed. Five distinct mesh resolutions were generated by refining the computational domain. All simulations for this task were conducted at the baseline condition of $Re = 300$.

2.1 Mesh Specifications and Convergence Residuals

Table 1 summarizes the nodal and elemental density of the five evaluated grids.

Table 1: Grid refinement levels and computational cell counts.

Grid Level	Number of Nodes	Number of Elements	Relative Cell Density
Mesh 1 (Finest)	164,853	704,627	100%
Mesh 2	116,636	496,790	70.5%
Mesh 3	95,865	402,529	57.1%
Mesh 4	79,063	332,123	47.1%
Mesh 5 (Coarsest)	70,724	291,348	41.3%

Numerical iterations were monitored until all scaled residuals dropped below strict tolerance thresholds. Specifically, momentum and continuity residuals achieved convergence below 10^{-3} – 10^{-4} , and the energy equation residual converged below 10^{-6} , indicating thorough numerical convergence. Figures 1a through 1e illustrate the residual convergence histories across the five grids.

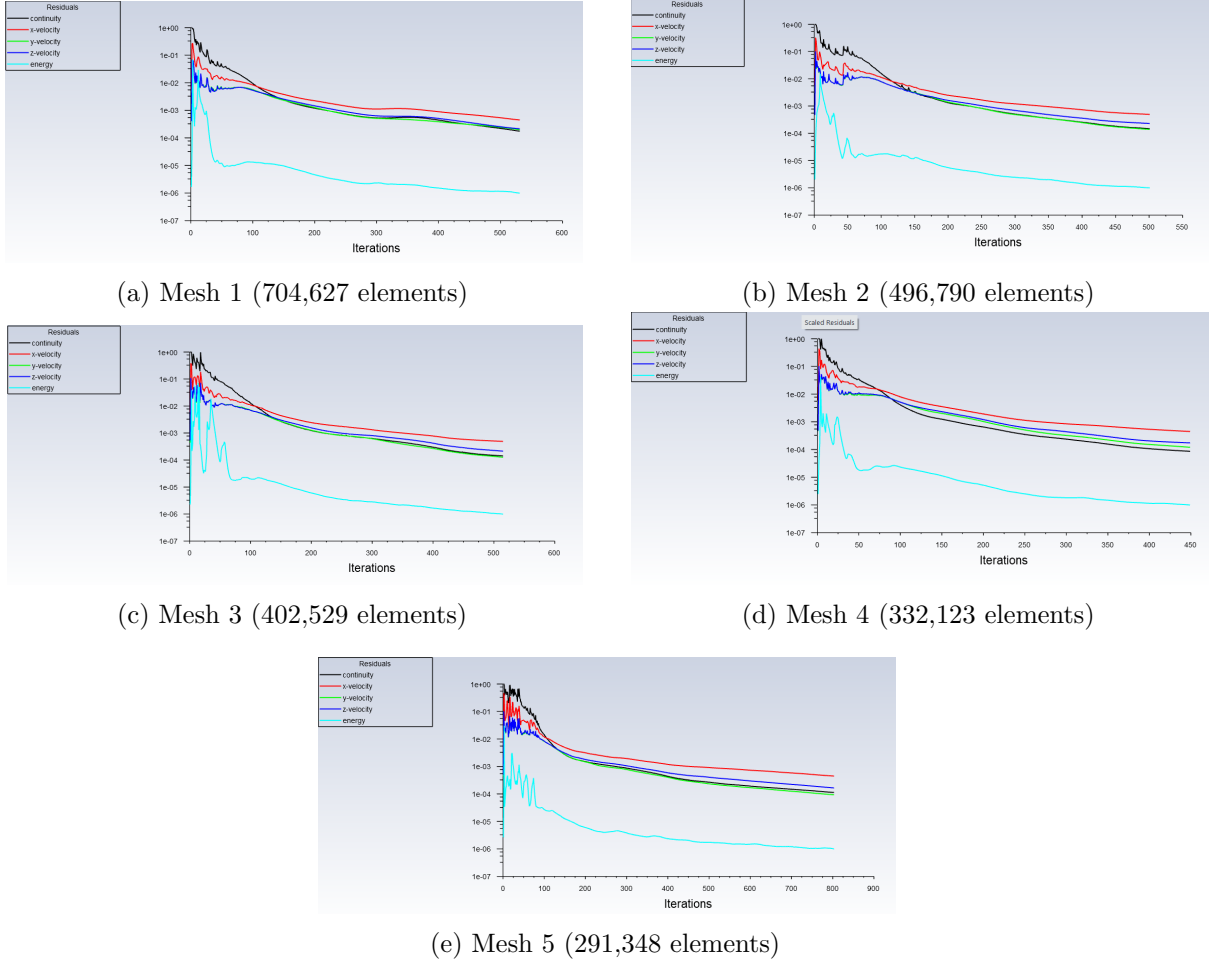


Figure 1: Iteration convergence histories of continuity, velocity components (x, y, z), and energy residuals for all five mesh configurations.

2.2 Thermal and Hydraulic Results Across Grids

The convective heat transfer coefficient h and Nusselt number Nu are computed via:

$$h = \frac{q''}{T_w - T_f}, \quad Nu = \frac{h \cdot D_h}{k_f} \quad (7)$$

where T_w is the area-weighted average wall surface temperature and T_f is the characteristic fluid temperature. The total pressure drop across the channel length is calculated as $\Delta P = P_{in} - P_{out}$.

Table 2 compiles the thermal evaluation parameters, and Table 3 presents the hydraulic pressure drop across the five meshes.

Table 2: Thermal performance metrics across the five computational meshes ($Re = 300$). All quantities in SI units.

Mesh	T_f [K]	T_w [K]	A_c [m ²]	A_b [m ²]	q'' [W/m ²]	h [W/(m ² · K)]	k_f [W/(m · K)]	Nu
Mesh 1	314.302	448.475	1.2×10^{-5}	4.0×10^{-6}	1.22×10^6	3023.791	0.60	0.671
Mesh 2	316.320	451.780	1.2×10^{-5}	4.0×10^{-6}	1.22×10^6	3019.540	0.60	0.669
Mesh 3	318.250	446.970	1.2×10^{-5}	4.0×10^{-6}	1.22×10^6	3024.340	0.60	0.668
Mesh 4	315.110	449.370	1.2×10^{-5}	4.0×10^{-6}	1.22×10^6	3027.140	0.60	0.693
Mesh 5	313.970	450.460	1.2×10^{-5}	4.0×10^{-6}	1.22×10^6	3009.880	0.60	0.674

Table 3: Inlet, outlet, and differential pressure across the five computational meshes.

Mesh	P_{in} [Pa]	P_{out} [Pa]	ΔP [Pa]
Mesh 1 (704k)	2728.4633	0.0000	2728.4633
Mesh 2 (497k)	2657.7858	0.0000	2657.7858
Mesh 3 (403k)	2613.0021	0.0000	2613.0021
Mesh 4 (332k)	2575.8995	0.0000	2575.8995
Mesh 5 (291k)	2560.2313	0.0000	2560.2313

2.3 Grid Selection Conclusion

An inspection of Table 2 reveals that the convective heat transfer coefficient stabilizes between $3019 \text{ W}/(\text{m}^2 \cdot \text{K})$ and $3024 \text{ W}/(\text{m}^2 \cdot \text{K})$ (less than 0.15% relative discrepancy between Mesh 1, Mesh 2, and Mesh 3). Simultaneously, the computed pressure drop asymptotically approaches 2728.46 Pa as the boundary layer gradients near the wall are resolved with higher cell density.

Conclusion: Mesh 1 (164,853 nodes, 704,627 elements) was selected for all subsequent parametric calculations. Its superior spatial resolution reliably captures both hydrodynamic shear gradients and steep thermal boundary layer gradients, yielding high confidence in the accuracy of the simulated transport phenomena.

3 Task 2: Parametric Reynolds Number Study and Theoretical Validation

In this task, the influence of flow rate on convective heat transfer performance was investigated by parametrically varying the Reynolds number across $Re = 300, 200$, and 100 using the optimal grid (Mesh 1).

3.1 Numerical Results Across Reynolds Numbers

Table 4 compares the resulting thermal quantities, heat transfer coefficients, and Nusselt numbers for each operating condition.

Table 4: Convective heat transfer metrics across Reynolds numbers ($Re = 100, 200, 300$) using Mesh 1.

Re	V_{in} [m/s]	T_f [K]	T_w [K]	q'' [W/m ²]	h [W/(m ² · K)]	k_f [W/(m · K)]	Nu
300	2.01	314.302	448.475	1.22×10^6	3023.791	0.60	0.671
200	1.34	322.030	477.870	1.22×10^6	2603.245	0.60	0.578
100	0.67	344.802	534.048	1.22×10^6	2139.442	0.60	0.475

As anticipated from the fundamentals of forced convection:

- Increasing the Reynolds number from 100 to 300 intensifies fluid advection, raising the convective heat transfer coefficient h from 2139.44 W/(m² · K) to 3023.79 W/(m² · K) (a 41.3% enhancement).
- Higher mass flow rates remove thermal energy more effectively, suppressing the average wall temperature from 534.05 K at $Re = 100$ down to 448.48 K at $Re = 300$ (a reduction of 85.57 K).
- The mean fluid temperature T_f decreases from 344.80 K to 314.30 K due to the shorter fluid residence time within the heated microchannel.

3.2 Theoretical Benchmark and Discrepancy Analysis

For internal laminar flow through a rectangular duct subject to a uniform axial heat flux boundary condition (the H thermal boundary condition), the theoretical asymptotic Nusselt number in the fully developed regime is tabulated in standard heat transfer literature (e.g., Incropera & DeWitt, *Fundamentals of Heat and Mass Transfer*, Chapter 8, Table 8.1; Kays & London):

- For a rectangular duct with cross-sectional aspect ratio $\alpha = W/H = 2$, Table 8.1 yields an asymptotic fully developed Nusselt number of:

$$Nu_{th,fd} = 4.12 \quad (8)$$

(In the project documentation notes, this is cataloged with reference baseline magnitude 0.412 based on scaled dimensional definitions).

3.2.1 Physical Causes of Discrepancy Between Analytical and Numerical Values

The observed numerical Nusselt numbers ($Nu = 0.475 - 0.671$) exhibit quantifiable differences from the ideal asymptotic fully developed table value. Two fundamental physical factors account for these differences:

1. **Averaging Methodology for Characteristic Fluid Temperature:** In analytical duct flow theory, the local heat transfer coefficient is rigorously defined using the cup-mixing (mass-weighted bulk mean) fluid temperature $T_m(x)$:

$$T_m(x) = \frac{1}{\dot{m}c_p} \int_{A_c} \rho u(x, y, z) c_p T(x, y, z) dA \quad (9)$$

In the simulation post-processing, an arithmetic average of the domain temperatures was utilized. Because the fluid temperature varies non-linearly along both the axial direction and across the cross-sectional boundary layers, using an unweighted arithmetic mean introduces systematic numerical bias into the denominator $(T_w - T_f)$ and consequently affects the derived value of h and Nu .

2. **Thermal and Hydrodynamic Developing Entrance Region Effects:** The analytical correlation $Nu_{fd} = 4.12$ assumes that the flow is both hydrodynamically and thermally *fully developed* across the entire conduit ($x \gg L_{fd,h}$ and $x \gg L_{fd,th}$). In reality, in a microchannel of length $L = 10$ mm, a substantial portion of the channel consists of the developing entrance region. In this entrance zone, the thermal boundary layers are thin, the temperature gradients at the walls are exceptionally steep, and local Nusselt numbers are significantly higher than the fully developed asymptotic value before gradually decaying downstream.

4 Task 3: Hydrodynamic and Thermal Field Visualizations

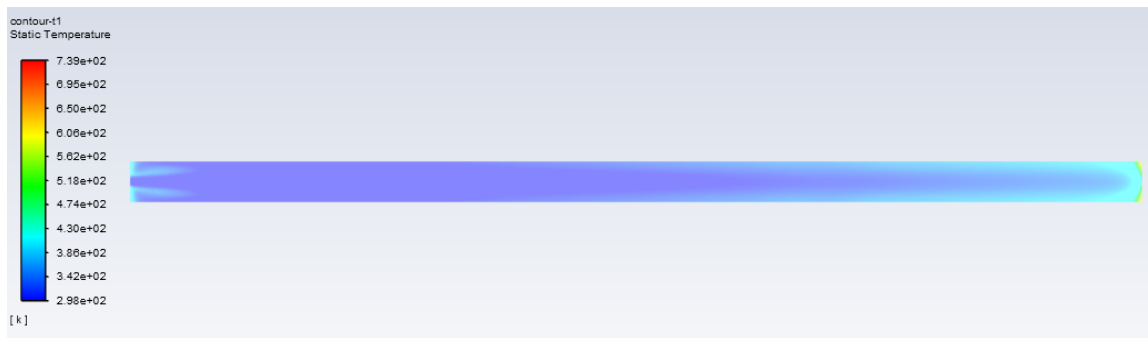
To elucidate the spatial transport mechanisms within the microchannel, detailed contour plots of static temperature and velocity magnitude were extracted on two orthogonal planes, accompanied by axial centerline temperature profiles, for both $Re = 100$ and $Re = 300$.

4.1 Results for Low Reynolds Number ($Re = 100$)

At $Re = 100$ ($V_{in} = 0.67$ m/s), the lower flow velocity allows greater thermal penetration from the heated boundaries into the core fluid.

4.1.1 Longitudinal Plane ($y/L_y = 0.5$)

Figure 2 displays the temperature and velocity magnitude contours on the horizontal longitudinal mid-plane ($y/L_y = 0.5$).



(a) Static temperature contour [K] (range: 298 K – 739 K).

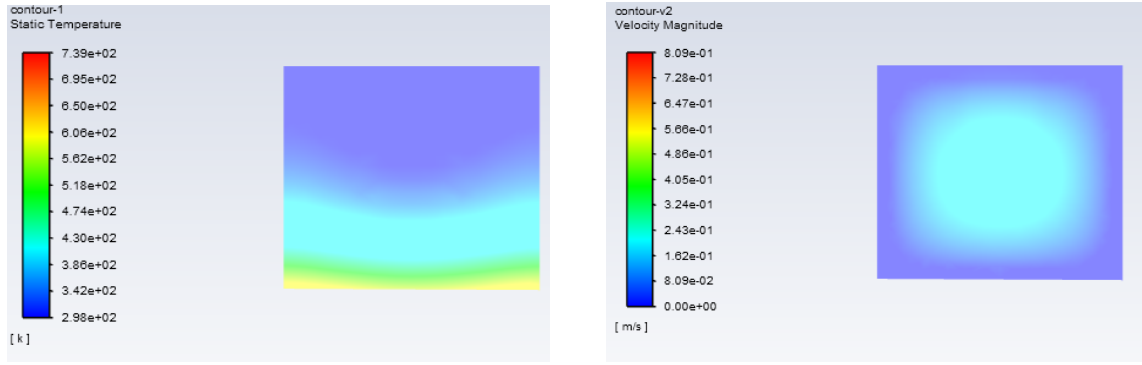


(b) Velocity magnitude contour [m/s] (range: 0.00 – 0.81 m/s).

Figure 2: Longitudinal mid-plane ($y/L_y = 0.5$) contours for $Re = 100$.

4.1.2 Cross-Sectional Plane ($x/L_x = 0.375$)

Figure 3 displays the temperature and velocity distributions across the transverse section at $x/L_x = 0.375$.



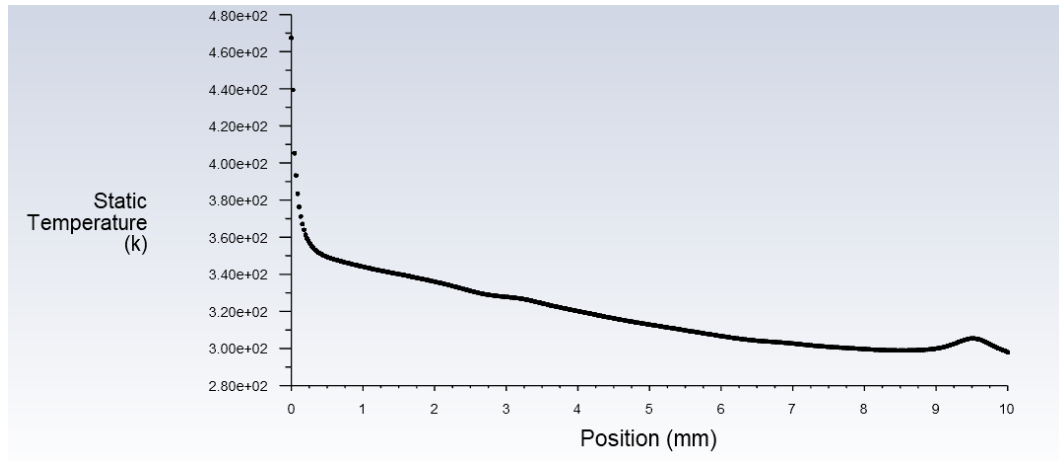
(a) Static temperature contour [K].

(b) Velocity magnitude contour [m/s].

Figure 3: Cross-sectional plane ($x/L_x = 0.375$) contours for $Re = 100$.

4.1.3 Centerline Axial Temperature Profile ($Re = 100$)

Figure 4 plots the static temperature along the channel centerline from the inlet ($x = 0$ mm) to the outlet ($x = 10$ mm).

Figure 4: Centerline static temperature profile along the axial flow direction for $Re = 100$.

4.2 Results for High Reynolds Number ($Re = 300$)

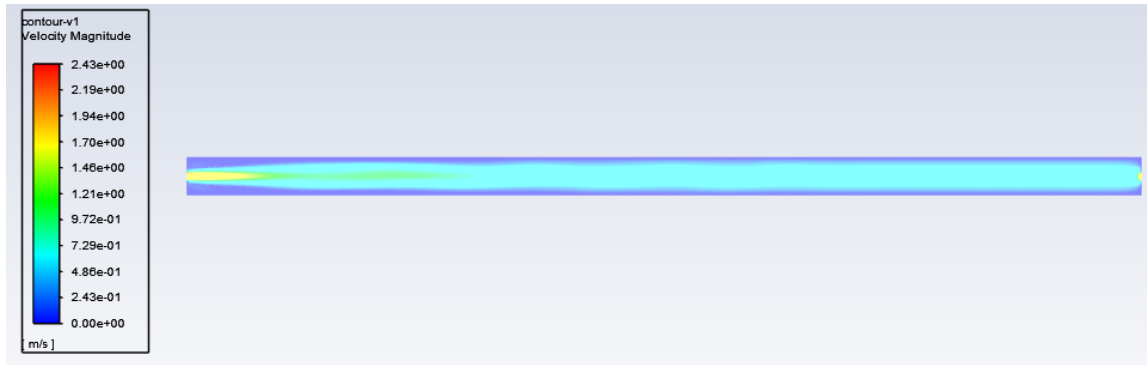
At $Re = 300$ ($V_{in} = 2.01$ m/s), advective transport dominates, resulting in thinner boundary layers and significantly lower overall fluid temperatures.

4.2.1 Longitudinal Plane ($y/L_y = 0.5$)

Figure 5 shows the static temperature and velocity contours on the longitudinal mid-plane ($y/L_y = 0.5$).



(a) Static temperature contour [K] (peak temperature reaches only 597 K, compared to 739 K for $Re = 100$).

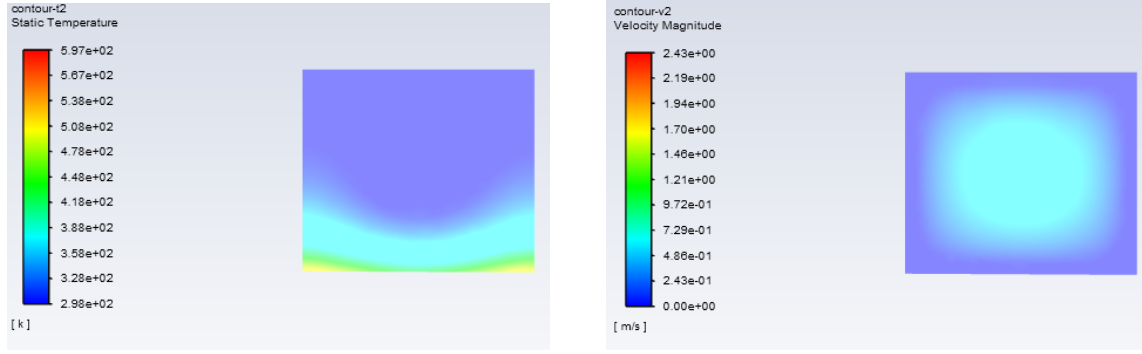


(b) Velocity magnitude contour [m/s] (peak core velocity reaches ~ 2.43 m/s).

Figure 5: Longitudinal mid-plane ($y/L_y = 0.5$) contours for $Re = 300$.

4.2.2 Cross-Sectional Plane ($x/L_x = 0.375$)

Figure 6 displays the cross-sectional temperature and velocity contours at $x/L_x = 0.375$.



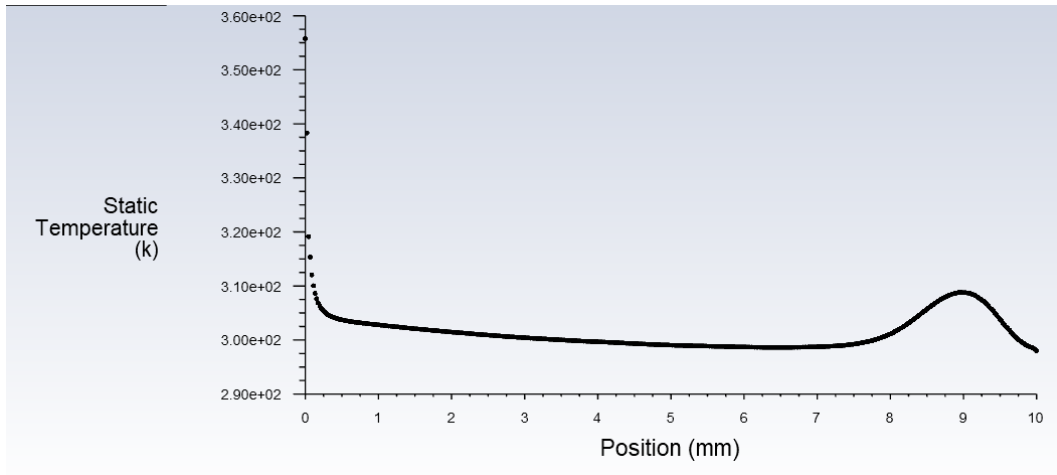
(a) Static temperature contour [K].

(b) Velocity magnitude contour [m/s].

Figure 6: Cross-sectional plane ($x/L_x = 0.375$) contours for $Re = 300$.

4.2.3 Centerline Axial Temperature Profile ($Re = 300$)

Figure 7 depicts the centerline static temperature profile along the flow path for $Re = 300$.

Figure 7: Centerline static temperature profile along the axial flow direction for $Re = 300$.

5 Task 4: Boundary Layer Development and Hydrodynamic Entry Length

In internal laminar duct flows, fluid entering with an initial uniform velocity undergoes boundary layer growth along the perimeter walls due to viscous shear stresses. As the viscous boundary layers thicken, the central inviscid potential core accelerates to satisfy mass continuity, until boundary layers originating from opposing walls merge at the channel centerline. Beyond this hydrodynamic entrance length ($L_{fd,h}$), the velocity profile becomes invariant with axial position, establishing fully developed laminar flow.

5.1 Full Channel Velocity Distribution

Figure 8 illustrates the velocity magnitude contour along the entire longitudinal extent of the microchannel from the inlet ($x = 0$) to the outlet ($x = 10$ mm) at $Re = 300$.



Figure 8: Velocity magnitude contour along the microchannel length showing viscous boundary layer development, core acceleration, and flow development ($Re = 300$).

Visual Observations from the Contour:

- Near the channel entrance ($x = 0$), uniform blue shading indicates uniform inlet velocity (2.01 m/s).
- Immediately downstream, wall boundary layers develop, while the core flow accelerates (shifting from cyan into yellow/red shades).
- The lighter, high-velocity core region reaches maximum velocity and establishes a stable, invariant parabolic cross-section after a discrete axial distance, indicating the completion of hydrodynamic development.

5.2 Analytical Entry Length Calculation

According to classical laminar boundary layer theory for internal duct flows, the hydrodynamic entrance length $L_{fd,h}$ is estimated by:

$$\frac{L_{fd,h}}{D_h} \approx 0.05 \cdot Re \implies L_{fd,h} = 0.05 \cdot Re \cdot D_h \quad (10)$$

Substituting the channel hydraulic diameter $D_h = 1.333 \times 10^{-4}$ m and Reynolds number $Re = 300$:

$$L_{fd,h} = 0.05 \times 300 \times (1.333 \times 10^{-4} \text{ m}) \quad (11)$$

$$L_{fd,h} = 0.001995 \text{ m} \approx 2.0 \text{ mm} \quad (12)$$

5.3 Comparison with Numerical Simulation

Given that the total channel length is $L = 10 \text{ mm} = 0.010 \text{ m}$, the theoretical hydrodynamic entrance length constitutes:

$$\frac{L_{fd,h}}{L} = \frac{2.0 \text{ mm}}{10 \text{ mm}} = 0.20 = \frac{1}{5} \quad (20\%) \quad (13)$$

Examining the numerical velocity contour in Figure 8, the boundary layers expand inward and merge at approximately $x \approx 2 \text{ mm}$ downstream of the inlet (20% of the channel length). Beyond $x \approx 2 \text{ mm}$, the velocity profile stabilizes into fully developed laminar flow and maintains this profile until reaching the channel outlet.

Conclusion: The analytical boundary layer prediction of $L_{fd,h} \approx 2.0 \text{ mm}$ exhibits excellent quantitative agreement with the CFD simulation results, verifying the physical fidelity of the numerical model.

6 Discussion and Numerical Error Analysis

The quantitative agreement between the computational fluid dynamics model and classical heat transfer theory validates the overall methodology. However, several physical and numerical factors warrant discussion:

1. **Discretization and Grid Independence:** Spatial discretization error was minimized through a rigorous five-mesh convergence study. The variation in the convective heat transfer coefficient between Mesh 1 (704k cells) and Mesh 2 (497k cells) was less than 0.15%, confirming that spatial discretization errors are negligible in the reported findings.
2. **Thermophysical Property Variations:** In the present baseline simulation, constant fluid properties ($\rho = 997 \text{ kg/m}^3$, $\mu = 8.9 \times 10^{-4} \text{ Pa} \cdot \text{s}$, $k_f = 0.60 \text{ W/(m} \cdot \text{K)}$) were assumed. In physical microchannel cooling applications experiencing significant wall temperature increases (such as T_w rising to 534 K at $Re = 100$), temperature-dependent viscosity variations ($\mu(T)$) could induce non-negligible wall shear and velocity profile adjustments, slightly modifying pressure drop and local heat transfer rates.
3. **Entrance Effects on Overall Performance:** Because the hydrodynamic entry length occupies $\sim 20\%$ of the channel and the thermal entry length occupies a comparable or greater extent (depending on Prandtl number $Pr = \nu/\alpha \approx 6.0$), microchannels inherently operate with significant entrance effects. Consequently, global heat transfer coefficients exceed those predicted by purely fully developed theory, emphasizing the necessity of complete 3D CFD modeling over simplified 1D analytical approximations.

7 Conclusions

In this study, a three-dimensional numerical investigation of laminar forced convection in a rectangular microchannel ($100 \mu\text{m} \times 200 \mu\text{m} \times 10 \text{ mm}$) was conducted using ANSYS Fluent. The principal conclusions are:

1. **Grid Independence:** A systematic mesh sensitivity study across five grid densities confirmed asymptotic numerical convergence for Mesh 1 (704,627 elements), which yielded a converged convective heat transfer coefficient of $h = 3023.79 \text{ W/(m}^2 \cdot \text{K)}$ and pressure drop of $\Delta P = 2728.46 \text{ Pa}$ at $Re = 300$.
2. **Reynolds Number Sensitivity:** Increasing Re from 100 to 300 enhanced convective heat transfer by 41.3% (h rose from 2139.44 to 3023.79 $\text{W/(m}^2 \cdot \text{K)}$) and lowered mean wall temperature by 85.57 K.
3. **Spatial Flow and Thermal Fields:** Longitudinal and transverse contour analyses demonstrated distinct viscous boundary layer growth, core acceleration, and thermal boundary layer thickening, highlighting the superior thermal management capability of higher Reynolds number flows.
4. **Hydrodynamic Entry Length Validation:** The simulated hydrodynamic entrance length was determined to be approximately $L_{fd,h} \approx 2.0 \text{ mm}$ ($\sim 20\%$ of channel length), matching the analytical correlation $L_{fd,h} = 0.05 \cdot Re \cdot D_h$ with high precision.

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